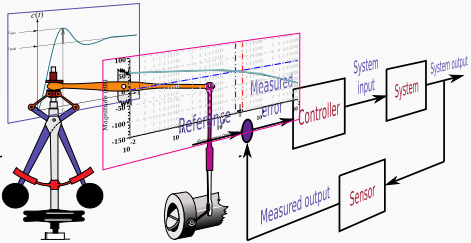
MAS508-G - Lecture 5 Control Design via Frequency Response

- Review of Frequency Response
- Gain Adjustment
- Lag Compensation
- Lead Compensation
- Lag-Lead Compensation
- PID vs. Lag-Lead Compensation

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Department of Engineering Sciences

$$G_c(s) = \frac{\left(s+\frac{1}{T_1}\right)\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{1}{\alpha T_1}\right)\left(s+\frac{\alpha}{T_2}\right)}$$



Outline

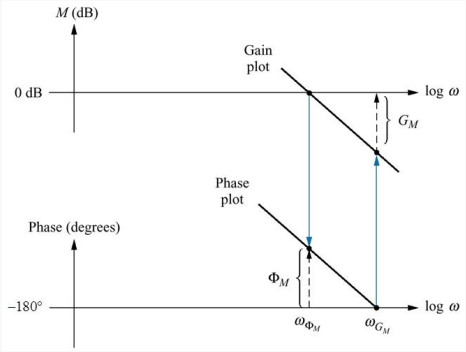
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Review of Frequency Response

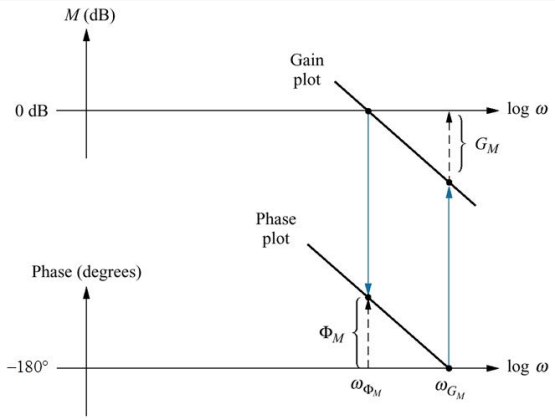
Review of Frequency Response

Gain Margin and Phase Margin

- Gain margin G_M: is the gain required to raise the magnitude curve to 0 dB.
  (The gain margin is the additional gain required to make the closed-loop system unstable.)
  - Phase plot: find the frequency ω_{G_M}, where the phase angle is $-180^o(+180^o)$.
  - Magnitude plot: determine the gain margin G_M at this frequency.
- Phase margin Φ_M: is the difference between the phase value and $-180^o(+180^o)$.
  (The phase margin is the additional phase required to make the closed-loop system unstable.)
  - Magnitude plot: find the frequency ω_{Φ_M} where the gain is 0 dB.
  - Phase plot: determine the phase margin ϕ_M at this frequency.



Stability via Frequency Response in Open-loop

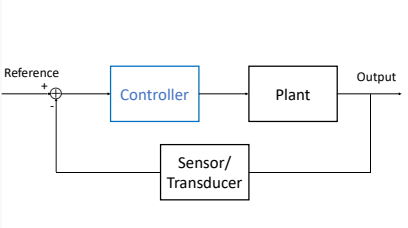


The system in closed-loop is stable if the magnitude in open-loop is less than 0 dB at the frequency where the phase is $\pm 180^o$.

The system in closed-loop is stable if G_M (in open-loop) is positive.

Assumptions

- Feedback system:



- Second Order Approximation: Assume dominant second-order closed-loop pole pair without zero, then the closed-loop transfer function

$$T(s) = \frac{C(s)}{R(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)H(s)} \approx \frac{K\omega_n^2}{s^2 + 2\zeta\omega s + \omega_n^2}$$

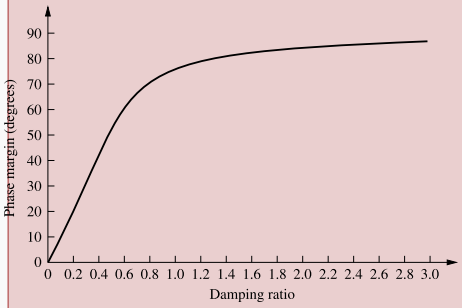


Estimate the time response performance of the closed-loop system $T(s)$ using open-loop frequency response of $C(s)G(s)H(s)$.

Overshoot via Open-loop Frequency Response

Relationship between damping ratio (equivalently percent overshoot) and phase margin:

Using **second order approximation:**



$$\phi_M = \tan^{-1} \frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{1+4\zeta^4}}}$$

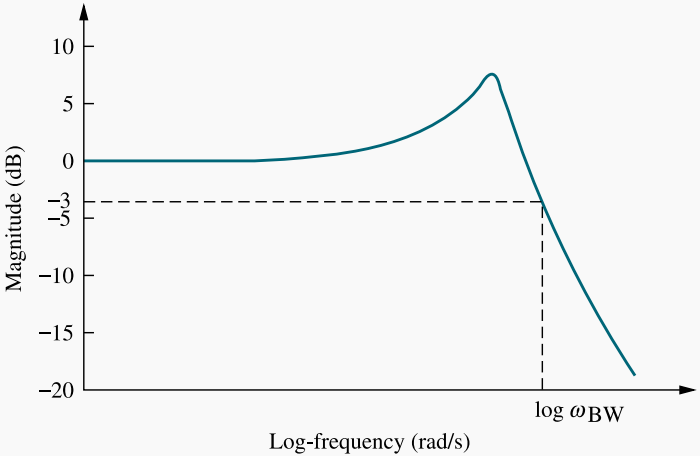
$$\zeta = \frac{\sin(\phi_M)}{2\sqrt{\cos(\phi_M)}}, \quad 0 < \phi_M < 90^o$$

$$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

where ϕ_M is the phase margin of the open-loop frequency response.
ζ is the damping ratio of the closed-loop transfer function.

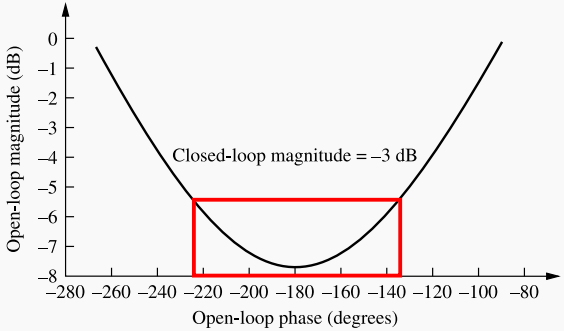
Bandwidth



Bandwidth of the closed-loop frequency response, defined as the frequency ω_{BW}:

ω_{BW} is the frequency at which the magnitude of the closed-loop frequency response is $3dB$ down from its value at zero frequency.

Bandwidth via Open-loop Frequency Response



Bandwidth ω_{BW}

- $\omega_{\mathbf{OL}|\mathbf{G_{dB}}\in[-\mathbf{7.5dB},-\mathbf{6dB}]}$ **and** $\Phi_{\mathbf{M}}\in[-\mathbf{225}^\circ,-\mathbf{135}^\circ]$
- **the frequency at which the open-loop magnitude response is between** -7.5**dB and** -6**dB and the phase response is between** -225° **and** -135°**.**

Bandwidth vs. Closed-loop Natural Frequency, Settling Time and Peak Time

Second order approximation:

Relationship between bandwidth and natural frequency

$$\omega_{BW} = \omega_n \sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

Relationship between bandwidth and settling time

$$\omega_{BW} = \frac{4}{T_s \zeta} \sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

Relationship between bandwidth and peak time

$$\omega_{BW} = \frac{\pi}{T_p \sqrt{1-\zeta^2}} \sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

Conclusion: Speed of the response is increased by increasing the bandwidth.

Bandwidth vs. Closed-loop Natural Frequency, Settling Time and Peak Time

Second order approximation:

Relationship between bandwidth and natural frequency

$$\omega_n = \frac{\omega_{BW}}{\sqrt{(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}}}$$

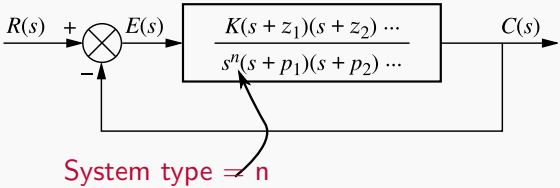
Relationship between bandwidth and settling time

$$T_s = \frac{4}{\omega_{BW}\zeta}\sqrt{(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}}$$

Relationship between bandwidth and peak time

$$T_p = \frac{\pi}{\omega_{BW}\sqrt{1-\zeta^2}}\sqrt{(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}}$$

Steady-state errors for Unity feedback systems



Input	Steady-state error formula	Type 0 Static error constant	Type 0 Error	Type 1 Static error constant	Type 1 Error	Type 2 Static error constant	Type 2 Error
Step, $u(t)$	$\frac{1}{1+K_p}$	K_p = Constant	$\frac{1}{1+K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	K_v = Constant	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	K_a = Constant	$\frac{1}{K_a}$

$K_p = \lim_{s\to 0} G(s)$, $K_v = \lim_{s\to 0} sG(s)$, $K_a = \lim_{s\to 0} s^2G(s)$, $G(s) = \frac{K(s+z_1)(s+z_2)\ldots}{s^n(s+p_1)(s+p_2)\ldots}$

Type $n \geq 3$ system: $K_p = \infty$, $K_v = \infty$, $K_a = \infty$, the steady-state error $= 0$ for unity feedback system with input as step, ramp and parabola.

Find Static Error Constants from Open-loop Frequency

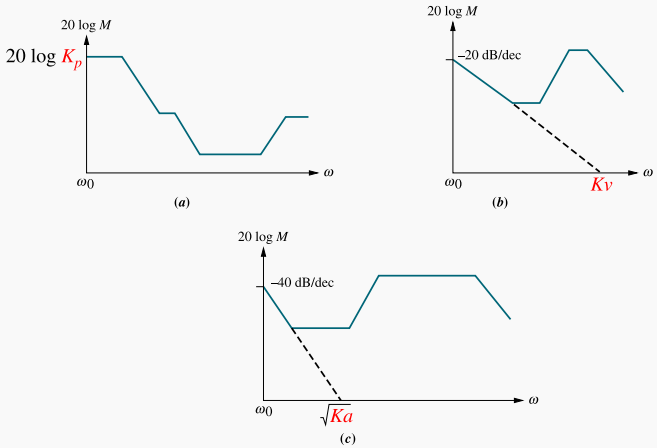


Figure: Typical unnormalized and unscaled Bode log-magnitude plots showing the value of static error constants **(a).** Type 0, **(b).** Type 1, **(c)**. Type 2 $G(s) = \frac{K(s+z_1)(s+z_2)\ldots}{s^n(s+p_1)(s+p_2)\ldots}$

Steady-state error is improved by increasing the Low Frequency magnitude responses, even if the High Frequency magnitude response is attenuated.

Key equations: Closed-loop Transient Response and Open-loop frequency

Second order approximation:

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + (\ln(\%OS/100))^2}} \tag{R.1}$$

$$\%OS = e^{-\left(\zeta\pi/\sqrt{1-\zeta^2}\right)} \times 100 \tag{R.2}$$

$$\phi_M = \tan^{-1}\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{1+4\zeta^4}}} \tag{R.3}$$

$$\zeta = \frac{\sin(\phi_M)}{2\sqrt{\cos(\phi_M)}}, \quad 0 < \phi_M < 90^o \tag{R.4}$$

$$\omega_{BW} = \omega_n\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} \tag{R.5}$$

$$\omega_{BW} = \frac{4}{T_s\zeta}\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} \tag{R.6}$$

$$\omega_{BW} = \frac{\pi}{T_p\sqrt{1-\zeta^2}}\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} \tag{R.7}$$

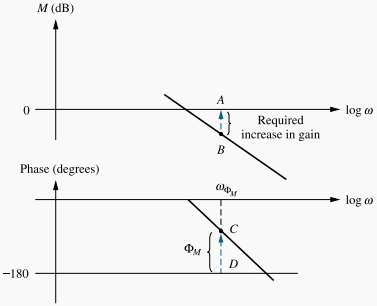
$$\omega_{BW} = \omega_{OL|G_{dB}\in[-6dB,-7.5dB]} \text{ if } \phi_{OL} \in [-135^\circ, -225^\circ] \tag{R.8}$$

Gain Adjustment

Book “Control Systems Engineering”, Chapter 11.2

Gain Adjustment Design Procedure: to satisfy the required $\%OS$ or phase margin

1. Draw the Bode plots of $G(s)$.
2. Using second order approximation, calculate the required phase margin ϕ_M from the percent overshoot, using equations (R.1) and (R.3).



3. Locate the frequency, ω_{ϕ_M} , on the Bode phase plot that yields the desired phase margin ϕ_M, CD.
4. Locate the gain AB on the Bode magnitude plot at the frequency ω_{ϕ_M}.
5. The adjust gain will satisfy $AB = 20\log(K)$ in order to yield the desired percent overshoot or phase margin. So the gain $K = 10^{AB/20}$.

Gain Adjustment Design Procedure: to satisfy the required steady-state error

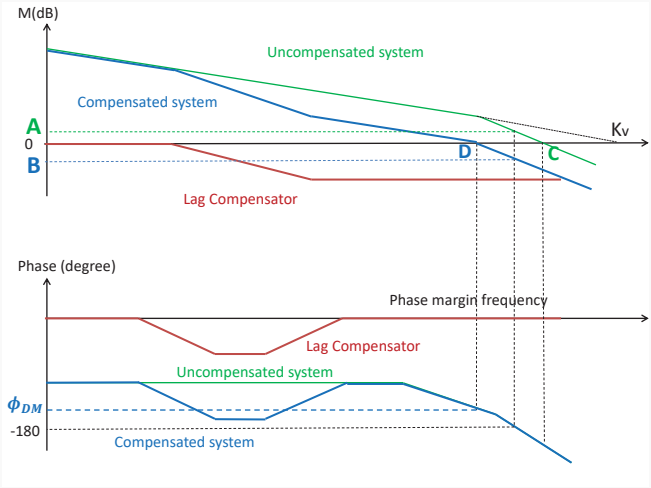
1. Draw the Bode plots of $G(s)$.
2. Determine the type of the system from Bode Plots.
3. Calculate the desired static error constants K_p, K_v or K_a from the required steady-state error.
4. Reading from Bode plots, find the gain to yield the desired the steady-state constant.
   - Type 0: Find the magnitude $M(0)$ of open-loop $G(s)$ at frequency $= 0$ rad/s.
     Calculate (K_{ess} using the equation $20\log(K_p) - M(0) = 20\log(K_{ess})$.
   - Type 1: The first asymptote is a straight line with slope -20 dB/dec.
     Find the magnitude M_{K_v} of the first asymptote at frequency $= K_v$ rad/s.
     Calculate (K_{ess} using the equation $0 - M_{K_v} = 20\log((K_{ess})$.
   - Type 2: The first asymptote is a straight line with slope -40 dB/dec.
     Find the magnitude $M_{\sqrt{K_a}}$ of the first asymptote at frequency $= \sqrt{K_a}$ rad/s.
     Calculate (K_{ess} using the equation $0 - M_{\sqrt{K_a}} = 20\log((K_{ess})$.

Lag Compensation

Book “Control Systems Engineering”, Chapter 11.3

Visualization Lag Compensation.

- Low frequency gain is not changed, without affecting the steady-state error.
- Reduce the cross-over frequency (from C to D), so the system becomes slower.
- Make the system stable (Gain margin from $A > 0$ to $B < 0$)

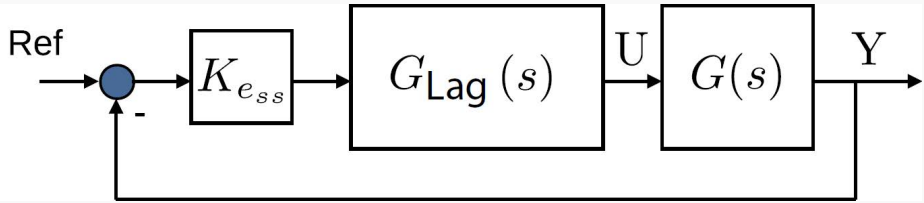


Lag Compensation

Goal: The following $G_{Lag}(s)$ will increase the phase margin (reduce the overshoot) without affecting the steady-state error.

$$G_{Lag}(s) = \frac{b}{a}\left(\frac{s+a}{s+b}\right), \quad 0 < b < a$$

- Lag compensator consists of zero $-a$ and pole $-b$.
- $b < a$.



Lag Compensation

$$G_{c_{\text{Lag}}}(s) = K_{ess} G_{\text{Lag}}(s) \tag{L.1}$$

$$G_{\text{Lag}}(s) = \frac{b}{a}\left(\frac{s+a}{s+b}\right), \quad b < a \tag{L.2}$$

$$a = 0.1\omega_p \tag{L.3}$$

$$b = a \times 10^{-\frac{G_{\text{adj}}}{20}} = 0.1\omega_p \times 10^{-\frac{G_{\text{adj}}}{20}} \tag{L.4}$$

where:

- $K_{e_{\text{ss}}}$: Gain adjustment in order to meet the required steady-state error.
- ω_p: Frequency at which the phase margin would be the desired one plus 5° to 12° extra.
- G_{adj}: Required Gain adjustment in dB to have 0 dB at $\omega = \omega_p$

$$G_{c_{\text{Lag}}} = K_{e_{\text{ss}}} 10^{-\frac{G_{\text{adj}}}{20}} \left(\frac{s+0.1\omega_p}{s+0.1\omega_p \times 10^{-\frac{G_{\text{adj}}}{20}}}\right) \tag{L.5}$$

Lag Compensator-Design Procedure

1. Set the gain K_{ess} to the value that satisfies the steady-state error specification.
   - If the transfer function of system $G(s)$ is known, calculate K_{ess}.
   - If only the Bode plots of $G(s)$ is known, find the gain K_{ess} using the relation between static error constant and Bode plot.

   (Then draw the Bode plot of the open-loop system with this gain K_{ess}.)
2. Calculate the desired phase margin ϕ_M to meet the percent overshoot requirement.
3. Find the frequency ω_p where the phase margin is 5° to 12° greater than the desired phase margin. (for example, $\phi_M + 10$ degree)
4. Find Gain adjustment G_{adj} in order to have 0 dB at this frequency ω_p rad/s.
   - Reading from Bode plots of $K_{ess}G(s)$ at ω_p: $G_{adj} = \text{Mag}_{K_{ess}G(s)}(\omega_p)$ dB
   - Reading from Bode plots of $G(s)$ at ω_p: $G_{adj} = \text{Mag}_{G(s)}(\omega_p) + 20\log(K_{ess})$ dB
5. Insert K_{ess}, ω_p, G_{adj} to Lag compensator.

$$G_{c_{\text{Lag}}} = K_{ess}10^{-\frac{G_{\text{adj}}}{20}}\left(\frac{s+0.1\omega_p}{s+0.1\omega_p \times 10^{-\frac{G_{\text{adj}}}{20}}}\right)$$

Key Equation

$$
\begin{aligned}
G_{c_{\text{Lag}}} &= K_{ess}\frac{b}{a}\left(\frac{s+a}{s+b}\right), \quad 0 < b < a \\
&= K_{ess}10^{-\frac{G_{\text{adj}}}{20}}\left(\frac{s+0.1\omega_p}{s+0.1\omega_p \times 10^{-\frac{G_{\text{adj}}}{20}}}\right)
\end{aligned}
$$

where:

$a = 0.1\omega_p$	The upper break frequency (rad/s)
$b = a \times 10^{-\frac{G_{\text{adj}}}{20}}$	The lower break frequency (rad/s)
K_{ess}	Gain adjustment in order to meet the required steady-state error.
ω_p	Frequency at which the phase margin would be the desired one plus 5° to 12° extra.
G_{adj}	Gain adjustment in dB in order to have 0 dB at $\omega = \omega_p$

Summary

- P-Controller is a good choice for Type 1/Step.
- For Type 0/Step or Type 1/Ramp , a P-Controller is not sufficient. We introduce a lag controller to increase the phase margin and reduce the steady-state error.
- The lag compensator has three parameters: K_{ess}, a, b

$$G_{c_{\text{Lag}}}(s) = K_{e_{ss}}\, G_{Lag}(s) = K_{ess}\, \frac{b}{a} \left(\frac{s+a}{s+b} \right), \quad b < a$$

or

$$G_{c_{\text{Lag}}}(s) = K_{ess} 10^{-\frac{G_{\text{adj}}}{20}} \left(\frac{s + 0.1\omega_p}{s + 0.1\omega_p \times 10^{-\frac{G_{\text{adj}}}{20}}} \right)$$

where $a = 0.1\omega_p$, $b = 0.1\omega_p \times 10^{-\frac{G_{\text{adj}}}{20}}$.

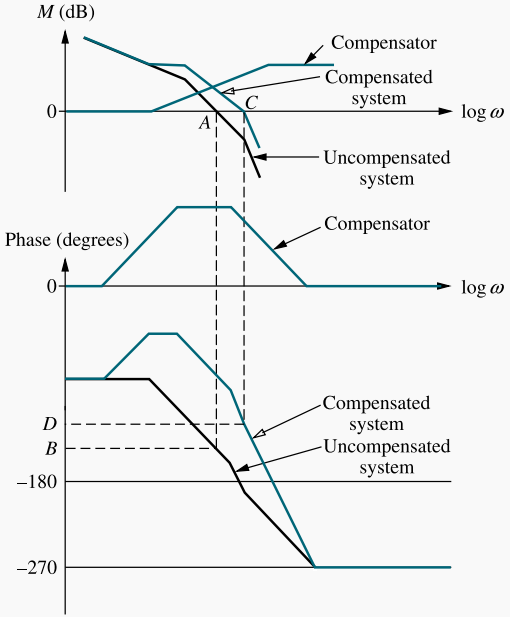
Summary-Lag Compensation

- Improve the steady-sate error. (steady-state error requirement)
- Improve the transient response. (overshoot requirement)
- Reduce the cross-over frequency and slow the system down.

$$G_{c_{\mathsf{Lag}}}(s) = K_{ess} G_{\mathsf{Lag}}(s)$$
$$G_{\mathsf{Lag}}(s) = \frac{b}{a}\left(\frac{s+a}{s+b}\right), \quad b < a$$

Lead Compensation Visualization

- Increase crossover frequency to realize a faster transient response (from A to C)
- Increase phase margin to reduce percentage overshoot (from B to D)

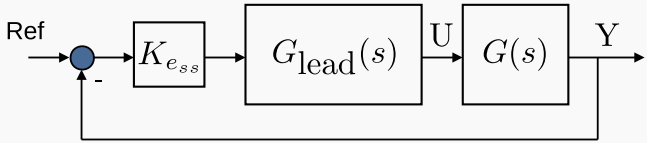


Lead Compensation

Book “Control Systems Engineering”, Chapter 11.4

Lead Compensation

Goal: Reduce percentage overshoot and realize a faster transient response.



$$G_{c_{\text{Lead}}}(s) = K_{ess} G_{\text{Lead}}$$

$$G_{\text{Lead}}(s) = \frac{1}{\beta} \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \qquad \beta < 1$$

Lead compensator consists of parameters β and T.
Alternatively, we can write Lead compenstor as

$$G_{\text{Lead}}(s) = \frac{b}{a} \left(\frac{s + a}{s + b} \right), \quad b > a$$

Design 1-Lead Compensator: Key Equations and Structure

Second order approximation:

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} \quad \text{(K.1)}$$

$$\phi_M = \tan^{-1}\frac{2\zeta}{\sqrt{-2\zeta^2 + \sqrt{1+4\zeta^4}}} \quad \text{(K.2)}$$

$$\omega_{BW} = \frac{4}{T_s\zeta}\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} \quad \text{(K.3)}$$

$$\omega_{BW} = \frac{\pi}{T_p\sqrt{1-\zeta^2}}\sqrt{(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}} \quad \text{(K.4)}$$

$$\omega_{BW} = \omega_{OL|G_{dB}\in[-6dB,-7.5dB]} \quad \text{if } \phi_{OL} \in [-135^\circ, -225^\circ] \quad \text{(K.5)}$$

Lead compensation:

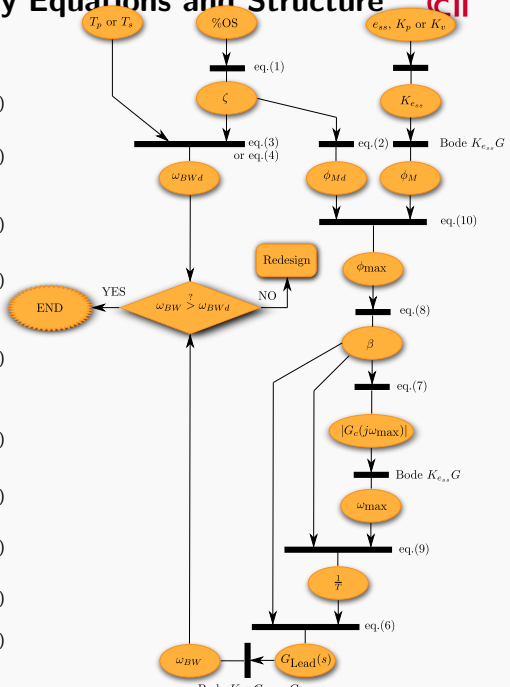
$$G_{\text{Lead}}(s) = \frac{1}{\beta}\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} \qquad \beta < 1 \quad \text{(K.6)}$$

$$|G_c(j\omega_{\text{max}})| = \frac{1}{\sqrt{\beta}} \quad \text{(K.7)}$$

$$\beta = \left(\frac{1-\sin\phi_{\text{max}}}{1+\sin\phi_{\text{max}}}\right) \quad \text{(K.8)}$$

$$\omega_{\text{max}} = \frac{1}{T\sqrt{\beta}} \quad \text{(K.9)}$$

$$\phi_{\text{max}} = \phi_{Md} - \phi_M + 10^\circ \quad \text{(K.10)}$$



Design 1: Lead Compensator with Steady-State and Phase Margin Requirements

1. Set the gain, K_{ess}, to satisfy the steady-state error requirement.
2. Re-Plot the Bode magnitude plot for this value of gain K_{ess} and determine the phase margin ϕ_M of uncompensated system $K_{ess}G(s)$.
3. Calculate the desired phase margin ϕ_{Md} to meet the percent overshoot requirement. Then evaluate the additional phase contribution required from the compensator. $\phi_{max} = \phi_{Md} - \phi_M + 10^o$ (add $[5^o \;\; 12^o]$)
4. Determine the value of $\beta = \left(\frac{1-\sin\phi_{\max}}{1+\sin\phi_{\max}}\right)$ (see Eq. (K.8)) from the lead compensator required phase contribution ϕ_{max}.
5. Determine the compensator magnitude $|G_c(j\omega_{\max})| = \frac{1}{\sqrt{\beta}}$ at the peak of the phase curve (see Eq. (K.7)). ($|G_c(j\omega_{\max})| = 20\log\left(\frac{1}{\sqrt{\beta}}\right)$ dB)
6. Determine $\omega_{\max}$ by finding where the uncompensated system $K_{ess}G(s)$ magnitude curve is the negative of $|G_c(j\omega_{\max})|$ dB. (Reading from bode plots of $K_{ess}G(s)$)
7. Find $\frac{1}{T} = \omega_{\max}\sqrt{\beta}$. (see Eq. (K.9))
8. Lead compensation (see Eq. (K.6)):

$$G_{C_{\text{Lead}}}(s) = K_{ess}\frac{1}{\beta}\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} \qquad \beta < 1$$

Design 2: Lead Compensator with Steady-State and Cross-Frequency Requirements (Increasing phase margin)

1. The maximum reachable phase margin is selected by choosing:

$$\omega_{\text{max}} = \omega_c \tag{K.11}$$

2. Choose a maximum phase lift $|G_c(j\omega_{\text{max}})|$ for the compensator such that the magnitude of the compensated system crosses 0 dB at frequency ω_{max}.

$$|G_c(j\omega_{\text{max}})| = 10^{|M(\omega_{\text{max}})|/20} \tag{K.12}$$

where $M(\omega_{\text{max}})$ is the magnitude in dB at $\omega = \omega_{\text{max}}$ reading from bode plot $K_{ess}G(s)$.

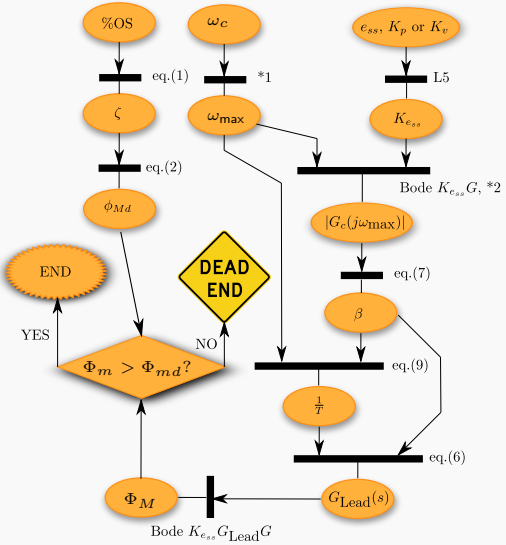
3. From equations (K.7) and (K.9)

$$\frac{1}{\beta} = |G_c(j\omega_{\text{max}})|^2 \tag{K.7}$$

$$\frac{1}{T} = \omega_{\text{max}}\sqrt{\beta} \tag{K.9}$$

4. Lead compensator:

$$G_{C_{\text{Lead}}}(s) = K_{ess}\frac{1}{\beta}\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} \qquad \beta < 1 \tag{K.6}$$



Summary-Lead Compensation

- Improve the steady-sate error. (steady-state error requirement)
- Improve the transient response. (overshoot requirement)
- Increase the cross-over frequency and realize a faster the system.

$$G_{c_{\text{Lead}}}(s) = K_{ess} G_{\text{Lead}}(s)$$
$$G_{\text{Lead}}(s) = \frac{b}{a}\left(\frac{s+a}{s+b}\right), \quad b > a$$

Lead and lag Compensation

Lag compensator:

$$G_{\text{Lag}}(s) = \frac{b}{a}\left(\frac{s+a}{s+b}\right), \qquad b < a$$
$$= \frac{1}{\alpha}\frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}} \qquad \alpha > 1$$

Lead compensator:

$$G_{\text{Lead}}(s) = \frac{b}{a}\left(\frac{s+a}{s+b}\right), \qquad b > a$$
$$= \frac{1}{\beta}\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} \qquad \beta < 1$$

Some conclusions:

- Lag compensator is good at reducing steady-state errors and increasing the stability but slow the system down.
- Lead compensator can speed up a system response, but do not increase the gain (stability) margin significantly.
- A slower system won't react as much to high frequency noise. Also it is safer if you are uncertain about your high frequency.
- Sometimes we need both: Lead-lag compensator combines the advantages of both the lag and the lead compensator.

	ϕ_M	G_M	Fast response (ω_{BW})
Lag Compensator	$++$	$++$	-
Lead Compensator	$+$	0^+	$++$
Lead-Lag Compensator	$+$	$+$	$+$

Lag-Lead Compensation

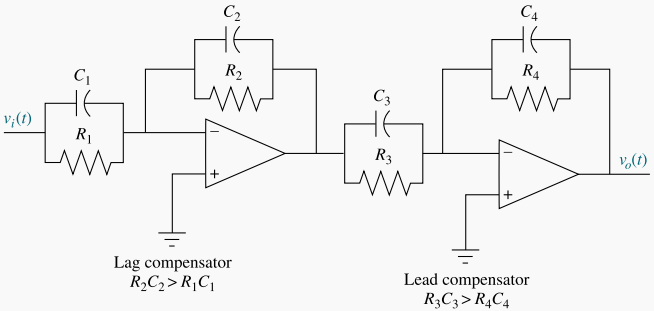
Book “Control Systems Engineering”, Chapter 11.5

Lag-Lead Compensation

$$
\begin{aligned}
G_c(s) &= K_{ess} G_{\text{Lead}}(s) G_{\text{Lag}}(s) \\
&= K_{ess} \underbrace{\gamma \left(\frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} \right)}_{G_{\text{Lead}}(s)} \underbrace{\frac{1}{\gamma} \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\gamma T_2}} \right)}_{G_{\text{Lag}}(s)} \\
&= K_{ess} \left(\frac{s + \frac{1}{T_1}}{s + \frac{\gamma}{T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\gamma T_2}} \right) \qquad \text{(G.1)}
\end{aligned}
$$

where $\gamma > 1$ and $T_1 \neq T_2$.

Active Circuit Lag-Lead Compensation



- Individual tuning of lag (T_1) and lead (T_2).
  $(G_{Lag} = -\frac{C_1}{C_2}\frac{s+\frac{1}{R_1C_1}}{s+\frac{1}{R_2C_2}}$, $G_{Lead} = -\frac{C_3}{C_4}\frac{s+\frac{1}{R_3C_3}}{s+\frac{1}{R_4C_4}})$
- First, design K_{ess} to satisfy steady-state error requirement.
- Second, design lag compensator in order to stabilise the system.
- Third, design lead compensator in order to satisfy the phase margin and bandwidth.
- Lag and Lead should operate in different frequency regions.

Lag-lead Compensator - Design Procedure

1. Using a second-order approximation, calculate the desired closed-loop bandwidth ω_{BW} to meet the settling time, peak time, or rise time requirement.
2. Set the gain, K_{ess}, to satisfy the steady-state error requirement.
3. Plot the Bode plots for this value of gain K_{ess}.
4. Using a second-order approximation, calculate the desired phase margin ϕ_M to meet the percent overshoot or damping ratio requirement.
5. Select a new phase-margin frequency $\omega_p = 0.8\omega_{BW}$ (cross frequency) (approximately 80% of the desired bandwidth).

Lag-lead Compensator - Design Procedure

6. Reading from Bode magnitude plot at the phase-margin frequency ω_p, determine the additional amount of phase lead required to meet the phase-margin requirement, $\phi(\omega_p)$. Add a small contribution ϕ_{add} (e.g. 10 degree) that will be required after the addition of lag compensator.

$$\phi_{\text{max}} = \phi_M + \phi_{add} - (180^o + \phi(\omega_p)) \tag{G.2}$$

7. Design the Lag compensator by selecting the higher break frequency at $\frac{1}{T_2} = 0.1\omega_p$. Calculate γ using following equation

$$\beta = \frac{1 - \sin(\phi_{\text{max}})}{1 + \sin(\phi_{\text{max}})} \quad (\phi_{\text{max}} = \arcsin\left(\frac{1-\beta}{1+\beta}\right)) \tag{G.3}$$

$$\gamma = \frac{1}{\beta} \tag{G.4}$$

$$\frac{1}{T_2} = 0.1\omega_p \tag{G.5}$$

$$\text{Lag: } G_{\text{lag}}(s) = \frac{1}{\gamma} \frac{\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{1}{\gamma T_2}\right)} \tag{G.6}$$

Lag-lead Compensator - Design Procedure

8. Design a lead compensator. Using the value of γ from the lag compensator design and the value assumed for the new phase-margin frequency, find the T_1 using the following equation

$$\frac{1}{T_1} = \omega_{\text{max}}\sqrt{\beta}, \quad (\omega_{\text{max}} = \omega_p) \tag{G.7}$$

$$\text{Lead: } G_{\text{lead}}(s) = \gamma \frac{\left(s + \frac{1}{T_1}\right)}{\left(s + \frac{\gamma}{T_1}\right)} \tag{G.8}$$

9. Check the final controller $G_c(s)$ such that the bandwidth (the speed requirement in Step 1) has been met.

$$\begin{aligned} G_c(s) &= K_{e_{ss}} G_{\text{lag}}(s) G_{\text{lead}}(s) \\ &= K_{e_{ss}} \left(\frac{\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{1}{\gamma T_2}\right)} \right) \left(\frac{\left(s + \frac{1}{T_1}\right)}{\left(s + \frac{\gamma}{T_1}\right)} \right) \end{aligned} \tag{G.9}$$

10. Redesign if phase margin or transient requirements are not met, as shown by analysis or simulation.

Lag-lead Compensator-Key equations

Lag-Lead compensator

$$\begin{aligned} G_{\mathsf{c}}(s) &= K_{e_{ss}} G_{\mathsf{lag}}(s) G_{\mathsf{lead}}(s) \\ &= K_{e_{ss}} \left(\frac{\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{1}{\gamma T_2}\right)} \right) \left(\frac{\left(s + \frac{1}{T_1}\right)}{\left(s + \frac{\gamma}{T_1}\right)} \right) \end{aligned} \tag{E.1}$$

Lag

$$G_{\mathsf{lag}}(s) = \frac{1}{\gamma} \frac{\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{1}{\gamma T_2}\right)} \tag{E.2}$$

$$\omega_p = 0.8\omega_{BW} \tag{E.3}$$

$$\phi_{\mathsf{max}} = \phi_M + \phi_{add} - (180^o + \phi(\omega_p)) \tag{E.4}$$

$$\frac{1}{T_2} = 0.1\omega_p \tag{E.5}$$

Lead

$$G_{\mathsf{lead}}(s) = \gamma \frac{\left(s + \frac{1}{T_1}\right)}{\left(s + \frac{\gamma}{T_1}\right)} \tag{E.6}$$

$$\beta = \frac{1 - \sin(\phi_{\mathsf{max}})}{1 + \sin(\phi_{\mathsf{max}})} \tag{E.7}$$

$$\gamma = \frac{1}{\beta} \tag{E.8}$$

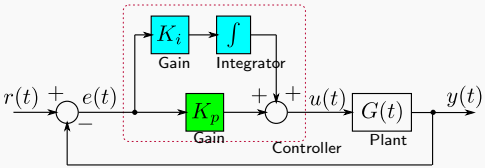
$$\frac{1}{T_1} = \omega_{\mathsf{max}} \sqrt{\beta} \quad (\omega_{\mathsf{max}} = \omega_p) \tag{E.9}$$

Summary

- Repetition of Lag and Lead Compensators.
- Introduction to Lag-Lead Compensator.
- Why Lag-Lead ??
  - Lag improves the phase margin, but reduces bandwidth.
  - Lead increases bandwidth but not significantly the phase margin.
  - Lag-Lead is a trade-of. The bandwidth is between lag and lead, improves steady-state errors and phase margin.
- You can design lag-lead as two separate controllers.
  - Select gain to satisfy steady-state error
  - Design lag to stabilize the system (not critical, any design for proper phase margin will be relegated to lead compensator)
  - Design lead to improve bandwidth and phase margin
  - Important: Lag and lead should operate in different frequency regions

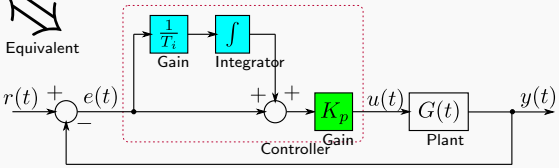
PID vs. Lag-Lead Compensator

PI



$$u(t) = K_i \int e(t)dt + K_p e(t)$$

Equivalent



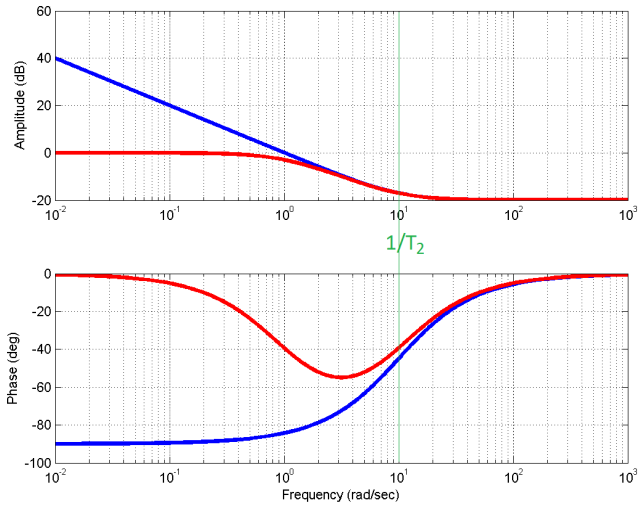
$$u(t) = K_p \left(\frac{1}{T_i} \int e(t)dt + e(t) \right)$$

where:

$$G_{PI} = K_p \frac{s + 1/T_i}{s}$$

PI vs. Lag Compensator : Bode Plot

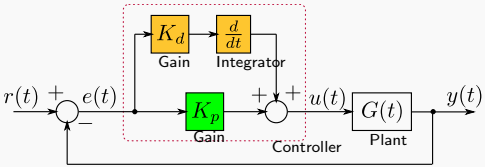
$$G_{PI} = \frac{1}{\alpha}\frac{s+\frac{1}{T_2}}{s}\ ,\ G_{Lag} = \frac{1}{\alpha}\frac{s+\frac{1}{T_2}}{s+\frac{1}{\alpha T_2}}\ ,\ \alpha > 1.$$



- Blue: PI
- Red: Lag
- PI is similar to lag.
- Keeping only break frequency $\frac{1}{T_2}$

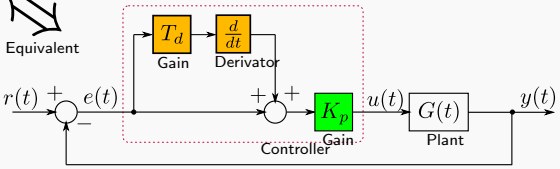
Anti-Windup PI = lag compensator!

PD



$$u(t) = K_d \frac{d\,e(t)}{dt} + K_p e(t)$$

Equivalent



$$u(t) = K_p \left(T_d \frac{d\,e(t)}{dt} + e(t) \right)$$

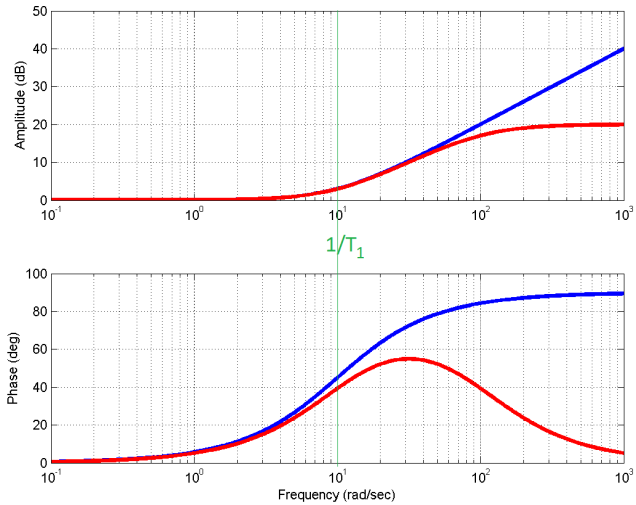
$$G_{PD} = K_p(T_d s + 1)$$

Replacing $\left\{ \frac{d}{dt} \right\} e(t) \rightarrow \left\{ \frac{\frac{d}{dt}}{1+N\frac{d}{dt}} \right\} e(t)$ $\qquad N = 10 - 20$

$$G_{PD} = K_p \frac{(T_d + N)s + 1}{Ns + 1}$$

PD Controller vs. Lead Compensator: Bode Plot

$G_{PD} = T_1\left(s + \frac{1}{T_1}\right)$, $G_{Lead} = \frac{1}{\beta}\frac{s+\frac{1}{T_1}}{s+\frac{1}{\beta T_1}}$, $0 < \beta < 1$.

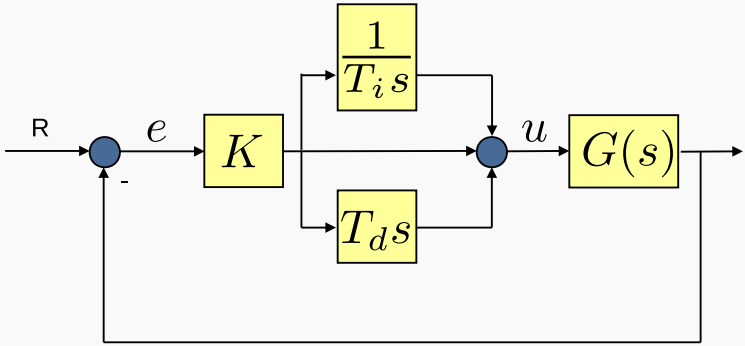


- Blue: PD
- Red: Lead
- PD is similar to Lead.
- Keeping only break frequency $\frac{1}{T_1}$

PD + filter = lead compensator!

PID Controller

$$u(t) = K\left(e(t) + \frac{1}{T_i}\int e(t)dt + T_d\frac{de}{dt}\right)$$



PID Controller Design

$$G_{\text{Lag-Lead}} = K_{\text{ess}} \frac{1}{\alpha\,\beta} \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\alpha\,T_2}} \right) \left(\frac{s + \frac{1}{T_1}}{s + \frac{1}{\beta\,T_1}} \right)$$

Use PI part as lag compensator.
Use PD part as lead compensator.

$$G_{\text{PID}}(s) = K \left(1 + \frac{1}{T_i s} + T_d s \right) = K \left(\frac{T_i s + 1 + T_d T_i s^2}{T_i s} \right)$$

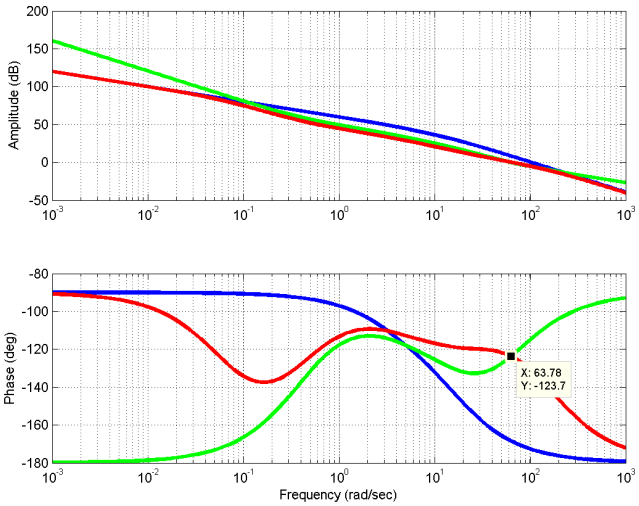
$\alpha > 1,\ 0 < \beta < 1$. Keeping only break frequencies $\frac{1}{T_2}$ and $\frac{1}{T_1}$, with same asymptote between them:

$$G_{\text{PID}}(s) = K_{\text{ess}} \frac{T_1}{\alpha} \frac{\left(s + \frac{1}{T_2}\right)\left(s + \frac{1}{T_1}\right)}{s} = K_{\text{ess}} \frac{T_1}{\alpha} \frac{\left(\frac{1}{T_1} + \frac{1}{T_2}\right) s + \frac{1}{T_1 T_2} + s^2}{s}$$

$$= \underbrace{K_{\text{ess}} \frac{T_1}{\alpha} \frac{1}{T_1 T_2}}_{=K/T_i} \frac{\overbrace{(T_1 + T_2)}^{=T_i} s + 1 + \overbrace{T_1 T_2}^{=T_i\,T_d}\, s^2}{s}$$



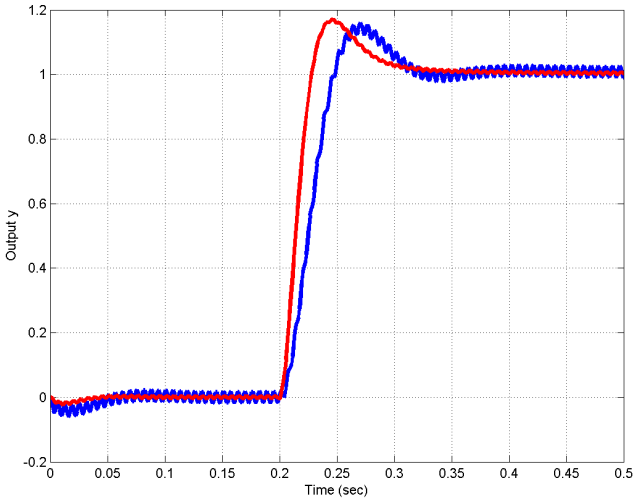
PID Controller vs. Lag-Lead Compensator: Bode Plot



- Blue: uncompensated
- Red: Lag-Lead
- Green: PID
- K gain in PID is chosen such that ω_c is the same as lag-lead compensator

In this case, almost the same phase margin.

Step response with noise in the measurements



- Blue: PID
- Red: Lag-Lead

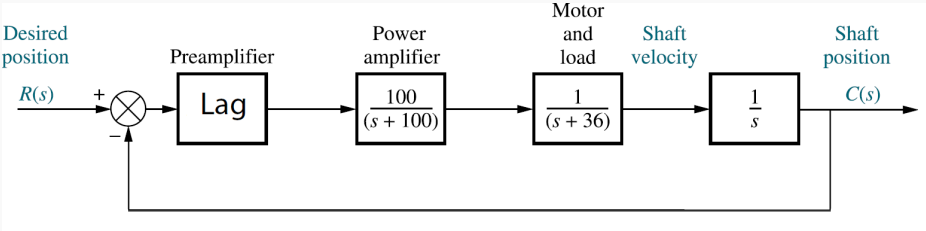
Why is the step response more noisy with a PID controller?

Summary

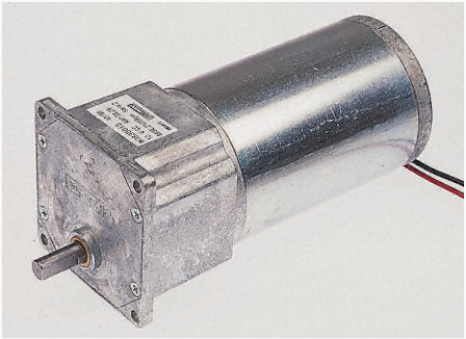
- PID vs. Lag-Lead
  - PI is similar to Lag Compensator
  - PD is similar to Lead Compensator
  - PID is similar to Lag-Lead Compensator
- Once a lag-lead compensator has been design, it can easily be translated into a PID controller.

Example 1: Lag-Compensation of DC Motor

Example 1: Lag-compensation of DC Motor



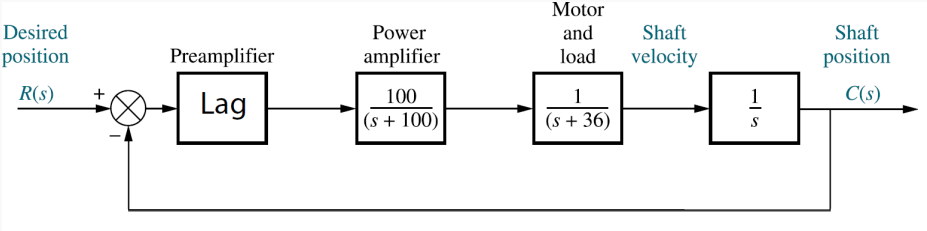
Question: Use Bode plots to design a lag-compensator to yield a 9.5% overshoot in the transient response for a step input and improvement in velocity constant to $K_v = 162.2$.



Lag-compensation of DC Motor: First step - Gain adjustment to satisfy K_v - Method 1

The open-loop transfer function with gain K_{ess} is $K_{ess}\frac{100}{s(s+100)(s+36)}$, the velocity constant is

$$\begin{aligned} K_v &= = \lim_{s\to 0}\left(sK_{ess}\frac{100}{s(s+100)(s+36)}\right) \\ &= \lim_{s\to 0}\left(K_{ess}\frac{100}{(s+100)(s+36)}\right) = \frac{K_{ess}}{36} = 162.2 \\ K_{ess} &= 162.2\times 36 = 5839 \end{aligned}$$



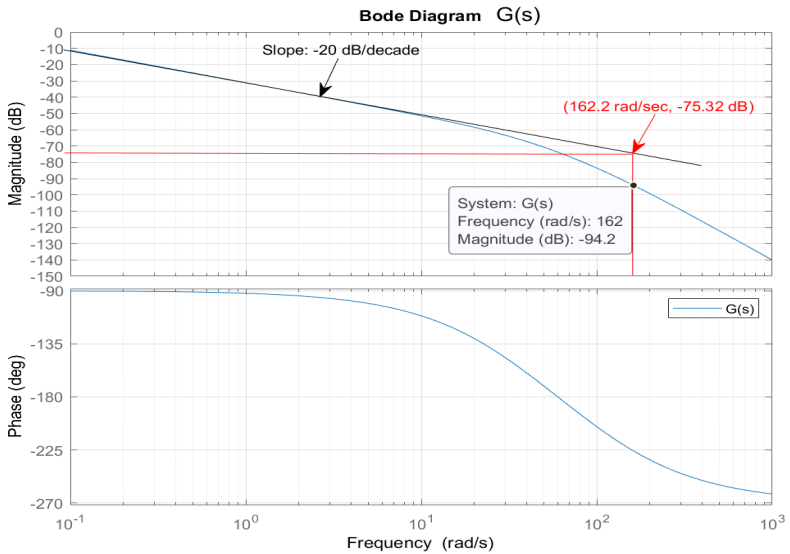
Lag-compensation of DC Motor: First step - Gain adjustment to satisfy K_v-Method 2

Method 2: The transfer function of $G(s)$ is unknown.

- Get the Bode plots of $G(s)$.
- Determine the type of the system from Bode Plots. (Type 1)
- Type 1: The first asymptote is a straight line with slope -20 dB/dec.
- Reading from Bode plots, find the magnitude $M_{K_v} \approx -75.32$ dB of the first asymptote at the frequency $\omega = K_v = 162.2$ rad/s, which equals to the desired static error constant $K_v = 162.2$.
- Calculate the gain K_{ess} using the equation.

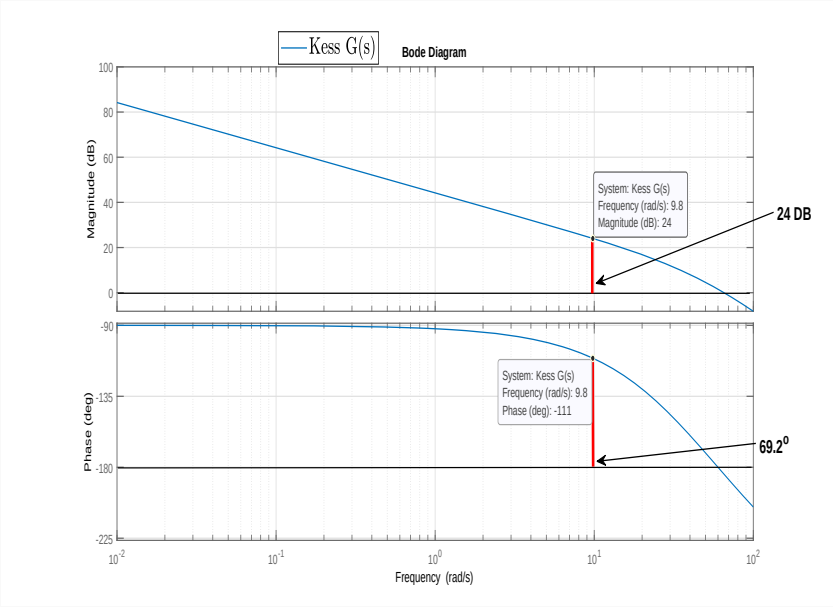
$$\begin{aligned} 0 - M_{K_v} &= 20\log(K_{ess}) \\ K_{ess} &= 10^{\frac{(0-M_{K_v})}{20}} \approx 10^{\frac{75.32}{20}} = 5835 \end{aligned}$$

Lag-compensation of DC Motor: First step - Gain adjustment to satisfy K_v-Method 2



Lag-compensation of DC Motor: Bode plots for $K_{ess} = 5839$

Gain adjustment system: $K_{ess}G(s) = 5839\frac{100}{s(s+100)(s+36)}$

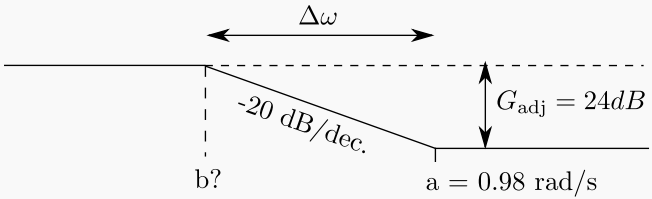


Lag-compensation of DC Motor: Second step

- 9.5% overshoot $\Leftrightarrow \zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2+\ln^2(\%OS/100)}} = 0.5996$. (see Eq. (R.1)).
- $\zeta = 0.5996$ gives the phase margin $\phi_M = \tan^{-1}\frac{2\zeta}{\sqrt{-2\zeta^2+\sqrt{1+4\zeta^4}}} = 59.2°$. (see Eq. (R.3))
- Increase this margin to $\phi_{Md} = 59.2° + 10° = 69.2°$ ($5°$-$12°$ extra)
- Find the frequency $\omega_p = 9.8$ rad/s where the phase angle of $(-180° + 69.2° = -110.8°)$. The magnitude at 9.8 rad/s is now $G_{adj} = 24$ dB. (At this frequency, the magnitude of the compensated system must go through 0 dB.)

Finding the Pole b of a Lag Compensator

- The high break-frequency $a = 0.1\omega_p = 0.98$ rad/s.
- Find the low break-frequency b.



$$\text{slope} = \frac{\Delta G}{\Delta\omega}$$

$$\Rightarrow -20 = \frac{-G_{\text{adj}} - 0}{\log(a) - \log(b)}$$

$$\Rightarrow \log(a) - \log(b) = \frac{-G_{\text{adj}}}{-20}$$

$$\Rightarrow \log(b) - \log(a) = -\frac{G_{\text{adj}}}{20}$$

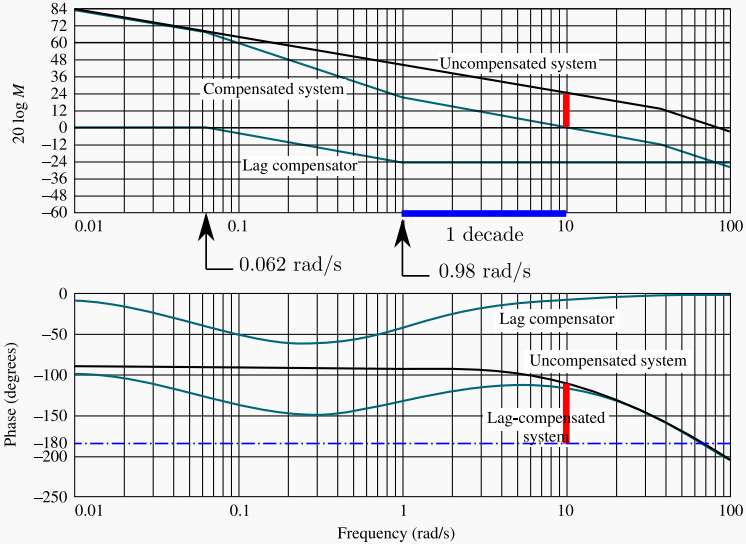
$$\Rightarrow b = a * 10^{-\frac{G_{\text{adj}}}{20}}$$

$$b = 0.98 * 10^{-\frac{24}{20}} = 0.062$$

Lag compensator:

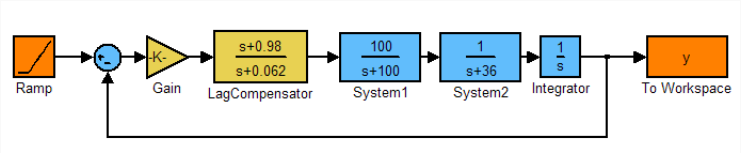
$$G_{c_{Lag}} = K_{ess}\frac{b}{a}\left(\frac{s+a}{s+b}\right) = 5839\frac{0.062}{0.98}\frac{s+0.98}{s+0.062}$$

Lag-compensation of DC Motor: Third step



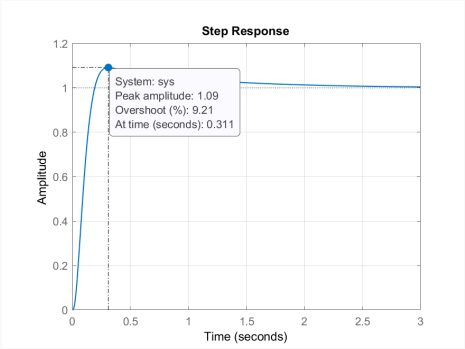
$$G_{c_{Lag}} = K_{ess}\frac{b}{a}\left(\frac{s+a}{s+b}\right) = 5839\frac{0.062}{0.98}\frac{s+0.98}{s+0.062}$$

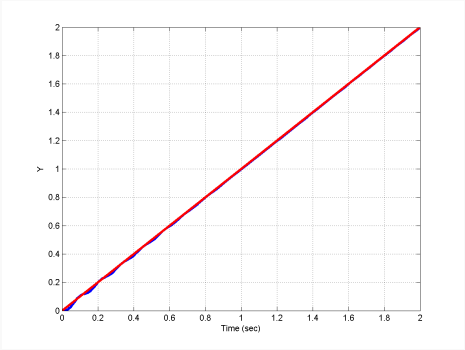
Simulation



$$K = K_{ess}\frac{b}{a} = 5839 \times \frac{0.062}{0.98}$$

Ramp response:





Characteristics of Lag-Compensated DC Motor

Tenfold improvement in steady-state error constant

Parameter	Proposed specification	Actual value
K_v	162.2	161.5
Phase margin	59.2°	63.7°
Phase-margin frequency	—	9.87 rad/s
Percent overshoot	9.5	9.2
Peak time	—	0.31 second

Overshoot remains almost same, because the phase margin.

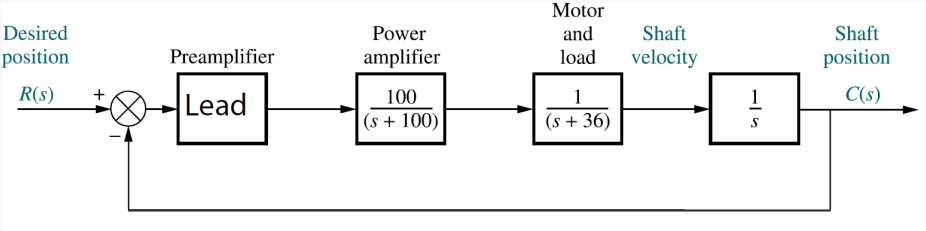
- Note the phase margin frequency! This is not a particularly fast controlled DC motor. But we have achieved the goal on overshoot and steady-state error.

To have a fast controlled DC motor is the next topic!

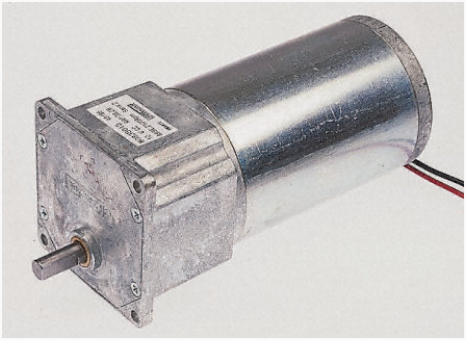
Example 2: Lead-Compensation of DC Motor

- Design 1: Overshoot requirement (phase marhin)
- Design 2: Speed requirement (cross-frequency)

Example 2: Lead-Compensation of DC Motor - Design 1



Question: Use Bode diagrams to design a lead-compensator to yield a 20% overshoot and $K_v = 40$, with a peak time of 0.1 second



Lead-compensation of DC Motor: Steps 1&2

- $K_v = 40$ and $G(s) = \frac{100}{s(s+36)(s+100)}$ yields:

$$K_v = \lim_{s\to 0} sKG(s) = \lim_{s\to 0} \frac{100K}{(s+36)(s+100)} = 40$$

$$K = K_{e_{ss}} = 40 \times 36 = 1440$$

- 20% overshoot $\Leftrightarrow \zeta = 0.456$.
  (see Eq. (K.1) $\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2+\ln^2(\%OS/100)}}$)

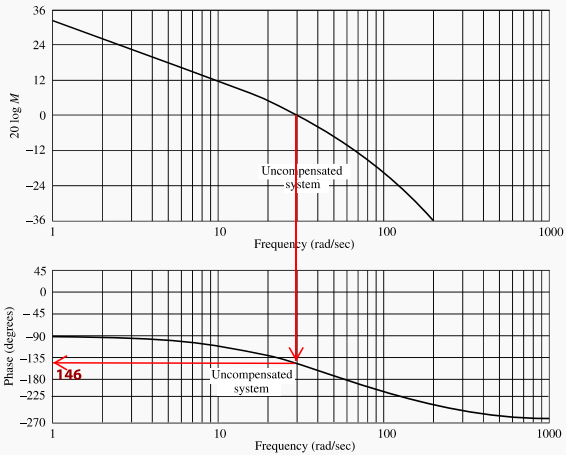
- $\zeta = 0.456$ gives $\phi_{Md} = 48.1°$ phase margin.
  (see Eq. (K.2) $\phi_M = \tan^{-1}\frac{2\zeta}{\sqrt{-2\zeta^2+\sqrt{1+4\zeta^4}}}$)

- $T_p = 0.1 \Leftrightarrow \omega_{BW} = 46.6$ rad/s
  (see Eq. (K.4) $\omega_{BW} = \frac{\pi}{T_p\sqrt{1-\zeta^2}}\sqrt{(1-2\zeta^2)+\sqrt{4\zeta^4-4\zeta^2+2}}$)

Lead-compensation of DC Motor: Steps 3&4

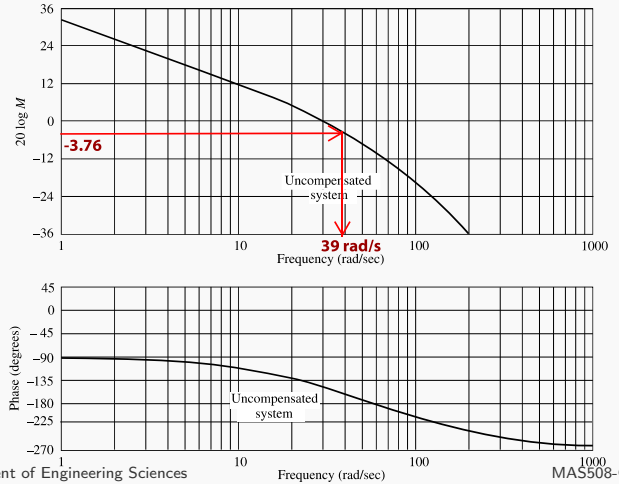
- 20% overshoot gives the desired phase margin $\phi_{Md} = 48.1^\circ$.

- Bode plot with $K = K_{ess} = 1440$
- From the bode plots, find the uncompensated phase margin $\phi_M = 34^\circ$ at $\omega_A = 29.6$ rad/sec. Since ω_{BW} will increase, we add an additional 10° margin. The additional phase contribution required from the Lead compensator is $\phi_{max} = \phi_{Md} - \phi_M + 10 = 48.1 - 34 + 10 = 24.1^\circ$.



Lead-compensation of DC Motor: Steps 5, 6, 7

- $\phi_{\text{max}} = 24.1^\circ$ yields $\beta = 0.42$. (see Eq. (K.8) $\beta = \left(\frac{1-\sin\phi_{\text{max}}}{1+\sin\phi_{\text{max}}}\right)$)
- $|G_c(j\omega_{\text{max}})| = \frac{1}{\sqrt{\beta}} = 1.54 = 20\log_{10}(1.54) = 3.76$ dB. (see Eq. (K.7))
- Find the frequency $\omega_{\text{max}} = 39$ rad/s, where the uncompensated system passes through $|-G_c(j\omega_{\text{max}}) = -3.76$ dB (step 7).

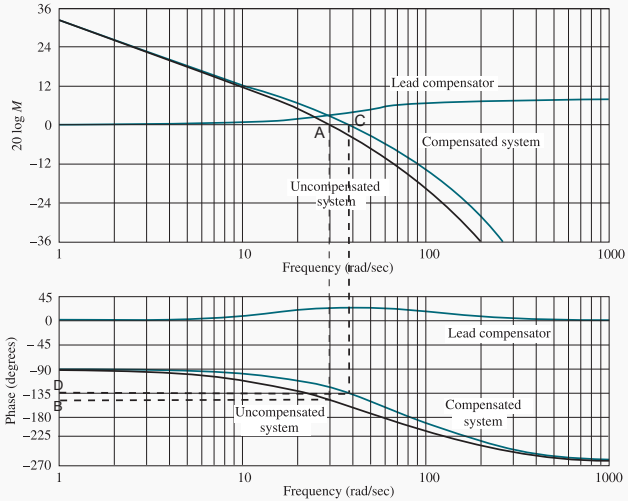


Lead-compensation of DC Motor: Steps 8 &9

- Thus, $\frac{1}{T} = 25.3$. (see Eq. (K.9) $\frac{1}{T} = \omega_{\text{max}}\sqrt{\beta}$)
- Lead:

$$\begin{aligned} & G_{\text{Lead}}(s) \\ = & \frac{1}{\beta}\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} \\ = & 2.38\frac{s+25.3}{s+60.2} \end{aligned}$$

- Lead-compensated system: $G_{C_{\text{Lead}}}(s) = K_{ess}G_{\text{Lead}}(s)$
- Finally Check bandwidth: $\omega_{BW} = 68.8$ rad/s



Characteristics of Lead-Compensated DC Motor

$$G_{C_{\text{Lead}}}(s) = K_{ess}\frac{1}{\beta}\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} = 2427\frac{s+25.3}{s+60.2}$$

Parameter	Proposed specification	Actual gain-compensated value	Actual lead-compensated value
K_v	40	40	40
Phase margin	48.1°	34°	45.5°
Phase-margin frequency	—	29.6 rad/s	39 rad/s
Closed-loop bandwidth	46.6 rad/s	50 rad/s	68.8 rad/s
Percent overshoot	20	37	22.6
Peak time	0.1 second	0.1 second	0.075 second

- Note the significant bandwidth improvements!

Example 2: Lead Compensation- Design 2

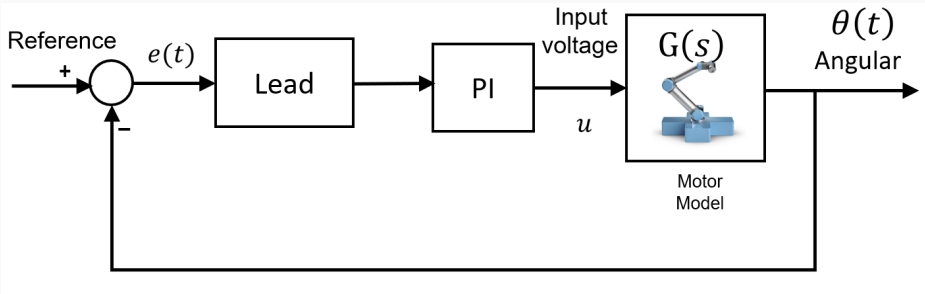


Figure: A robotic manipulator system

The system with PI control in open-loop: $PI(s)G(s)$

Lead Compensation Example - Design 2

The frequency response of the open-loop system is shown below.

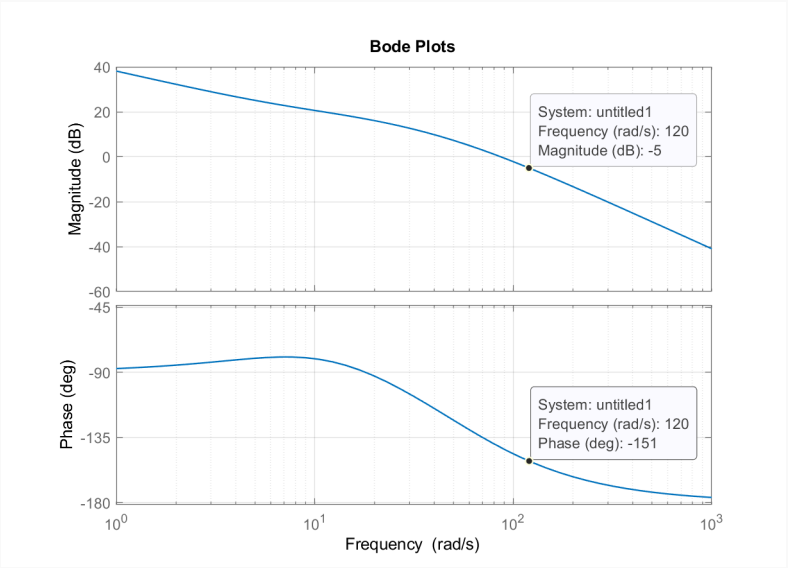


Figure: Frequency response of the system with PI control $PI(s)G(s)$

Lead Compensation Example 2- with Cross-Frequency Requirement

1. With the help of bode plots in previous slide, design a lead compensator to satisfy the speed requirement, while without affecting the steady-state error.
   - Speed requirement: the crossover frequency at 0dB for the compensated system in open-loop $\omega_c \geq 100$rad/s.

   Determine the values of β and T in the following Lead compensator.

$$G_{Lead}(s) = \frac{1}{\beta}\left(\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}}\right)$$

2. Is your Lead compensator designed in Part (1) able to fulfill the accuracy and stability requirements below? Explain your answer.
   - Stability requirement: Phase margin $\phi_M \geq 45^o$.
   - Accuracy requirement: Static position constant $K_p \geq 100$.

Lead Compensation Example 2- Design 2

1. Solution:
   - Desired speed requirement is the crossover frequency $\omega_c \geq 100$ rad/s.
   - We choose the frequency at the maximum phase is $\omega_{\text{max}} = \omega_c = 120$ rad/s.
   - Magnitude contribution required from the lead compensator in order to have 0 dB at $\omega_{\text{max}} = 120$rad/s (reading from Bode plots) :

$$|G_c(j\omega_{\text{max}})| = |-5|dB = 10^{5/20} = 1.7783$$

   - Corresponding β

$$|G_c(j\omega_{\text{max}})| = \frac{1}{\sqrt{\beta}}, \quad \Rightarrow \quad \beta = \frac{1}{|G_c(j\omega_{\text{max}})|^2} = \frac{1}{1.7783^2} = 0.3162$$

   - Phase contribution from the Lead compensator:

$$\beta = \left(\frac{1-\sin\phi_{\text{max}}}{1+\sin\phi_{\text{max}}}\right), \quad \Rightarrow \quad \phi_{\text{max}} = \arcsin\left(\frac{1-\beta}{1+\beta}\right) = 31.30^o$$

   - Time constant T of the Lead compensator:

$$\omega_{\text{max}} = \frac{1}{T\sqrt{\beta}}, \quad \Rightarrow \quad \frac{1}{T} = \omega_{\text{max}}\sqrt{\beta} = 67.48 sec^{-1}$$

   - Lead compensator is

$$G_{Lead}(s) = \frac{1}{\beta}\left(\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}}\right) = \frac{1}{0.3162}\left(\frac{s+67.48}{s+\frac{67.48}{0.3162}}\right) = 3.1626\frac{s+67.48}{s+213.41}$$

Lead Compensation Example 2- Design 2

1. Solution:

- The Lead compensator will not affect the initial asymptote of Bode plots. From the Bode plots, the system in open-loop is now of type 1. Therefore, the static position constant $K_p = \lim_{s\to 0} G_{lead}G_p = \infty$ is greater than 100 as required. So the accuracy requirement is satisfied.
- The phase of the system is -151° at 120 rad/s reading from the Bode plots. The phase margin of the system without lead is

$$\phi_M^{system} = -151^o - (-180^o) = 29^o$$

Adding the lead compensator to the system will bring the crossover frequency to 120 rad/s and add the phase contribution $\phi_{max} = 31.3^o$ at this same frequency. The resulting phase margin is therefore:

$$\phi_M = \phi_M^{system} + \phi_{max} = 29^o + 31.3^o = 60.3^o$$

which is larger than the stability requirement $\phi_M \geq 45^o$. So the stability requirement is satisfied.

Lead Compensation Example 2- Design 2

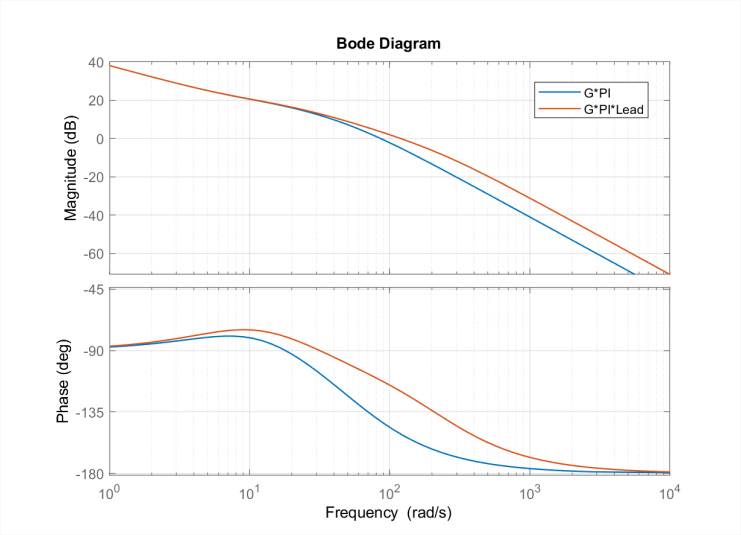


Figure: Bode plots of Lead compensated system

Lead Compensation Example 2- Design 2

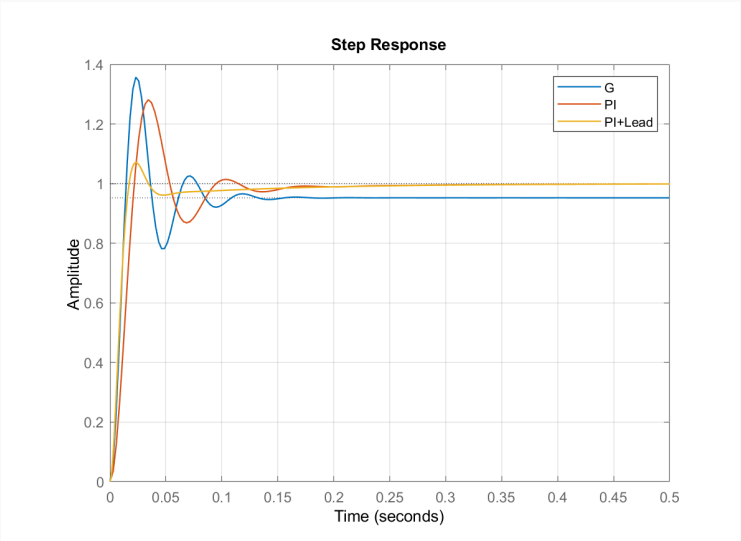
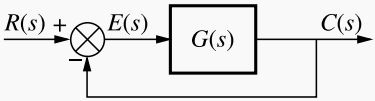


Figure: Step response

Example 3- Lag-Lead Compensation

Example 3: Lag-Lead Compensation

Given the system below (Nise Book 11.4):



where

$$G(s) = \frac{1}{s(s+1)(s+4)}$$

design a lag-lead compensator using Bode plots to yield a 13.25% overshoot, peak time of 2 seconds and K_v=12.

Example 3 -Step 1. find K_{ess} for steady-state error requirement $K_v = 12$

- 13.25% overshoot $\rightarrow \zeta = 0.541$, which yields $\phi_M = 55^\circ$
- $T_p = 2.0 \rightarrow \omega_{BW} = 2.29$ rad/s
- From $G(s) = \frac{1}{s(s+1)(s+4)}$ and the required velocity constant $K_v = 12$ yield

$$\begin{aligned} K_v &= \lim_{s=0} sK_{e_{ss}} G(s) = \lim_{s=0} sK_{e_{ss}} \frac{1}{s(s+1)(s+4)} = K_{e_{ss}} \frac{1}{4} = 12 \\ K_{e_{ss}} &= 12 * 4 = 48 \end{aligned}$$

Example 3- Bode plot for $K_{ess}G(s) = \frac{48}{s(s+1)(s+4)}$

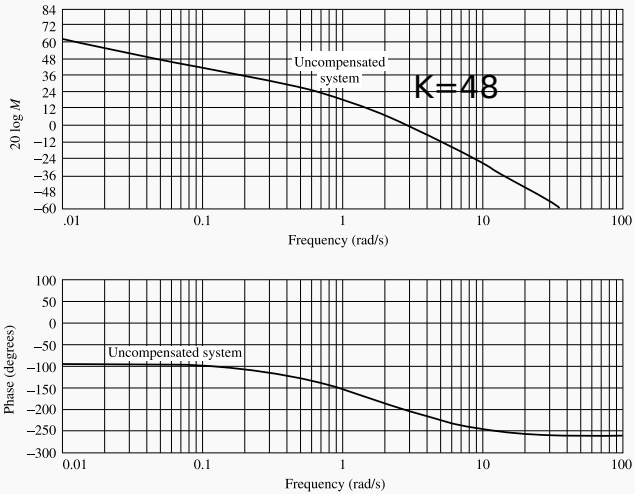


Figure: Bode plots for $K_{ess}G(s) = \frac{48}{s(s+1)(s+4)}$

- Select a new phase-margin frequency
  $\omega_p = 0.8\omega_{BW} = 2.29 * 0.8 = 1.83$ rad/s.
- At $\omega_p = 1.83$ rad/s, the phase angle is
  $\phi(\omega = \omega_p) = -176^o$.

Step 2: Lag Compensator

- Let us first design a lag compensator to stabilize the system.
- Calculate the maximum phase contribution from Bode plots and required phase margin.

$$\phi_{\text{max}} = \phi_M + 5^o - (180^o + \phi(\omega = \omega_p)) = 55 + 5 - (180 - 176) = 56^o$$

- Selecting the higher break frequency at

$$\frac{1}{T_2} = 0.1\omega_p = 0.1 \times 1.83 = 0.183$$

- Calculate the γ from

$$\beta = \frac{1 - \sin(\phi_{\text{max}})}{1 + \sin(\phi_{\text{max}})} = 0.094$$
$$\gamma = \frac{1}{\beta} = 10.6$$

- Lag compensator:

$$G_{\text{Lag}} = \frac{1}{\gamma} \frac{\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{1}{\gamma T_2}\right)} = \frac{1}{10.6} \frac{(s + 0.183)}{(s + 0.0172)}$$

Step 3: Design Lead Compensation

- Design a lead compensator. At the $\omega_{max} = \omega_p = 1.83$ rad/sec, using the values of β and γ from the lag compensator design, calculate the higher break frequency is

$$\frac{1}{T_1} = \omega_{\text{max}}\sqrt{\beta} = 1.83 \times \sqrt{0.094} = 0.56 \text{ rad/s}$$
$$\gamma = 10.6$$

- Lead compensator is

$$G_{\text{Lead}} = \gamma \frac{\left(s + \frac{1}{T_1}\right)}{\left(s + \frac{\gamma}{T_1}\right)} = 10.6 \frac{(s + 0.56)}{(s + 5.96)}$$

Final controller: Lag-Lead Compensation

Lag-Lead Compensator is

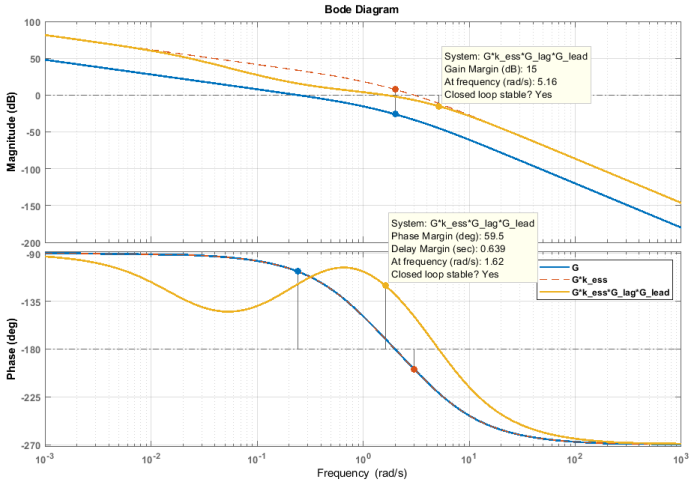
$$
\begin{aligned}
G_{C_{\text{Lag-Lead}}} &= K_{e_{ss}} G_{\text{lag}}(s) G_{\text{lead}}(s) \\
&= K_{e_{ss}} \left(\frac{\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{1}{\gamma T_2}\right)} \right) \left(\frac{\left(s + \frac{1}{T_1}\right)}{\left(s + \frac{\gamma}{T_1}\right)} \right) \\
&= 48 \left(\frac{(s + 0.183)}{(s + 0.0172)} \right) \left(\frac{(s + 0.56)}{(s + 5.96)} \right)
\end{aligned}
$$

Check the final Lag-Lead controller with Simulation.

Parameter	Proposed specification	Actual value
K_v	12	12
Phase margin	55°	59.3°
Phase-margin frequency	—	1.63 rad/s
Closed-loop bandwidth	2.29 rad/s	3 rad/s
Percent overshoot	13.25	10.2
Peak time	2.0 seconds	1.61 seconds

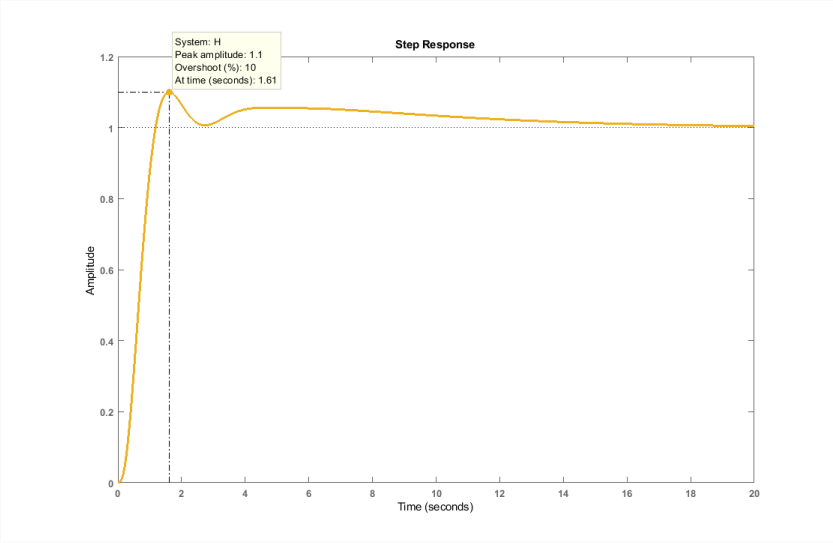
Characteristics of Lag-Lead Compensated System

Bode Plots: Lag-Lead compensation



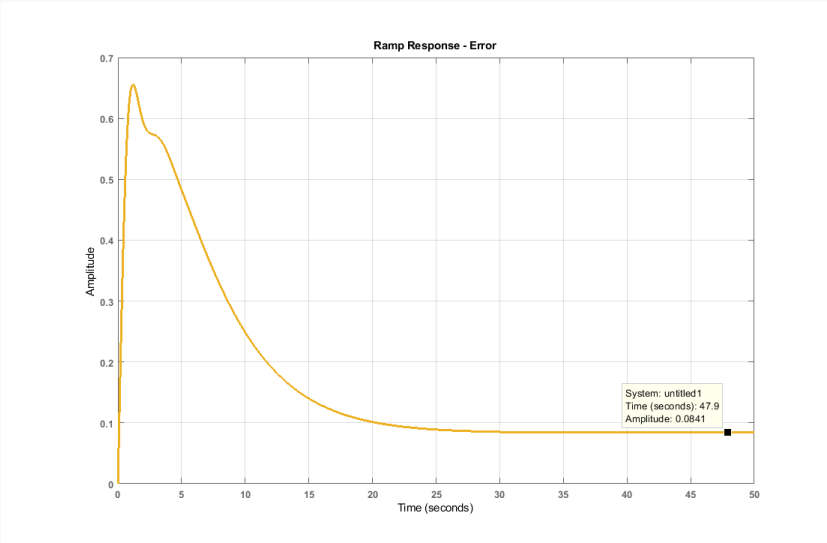
- Blue: System.
- Red: K_{ess}
- Yellow: K_{ess}-Lag-Lead
- $\omega_c = 1.6$ rad/s
- $\phi_M = 59.5°$
- $\%OS = 17.7$
- $T_p = 1.56$s.

Time response in Matlab: Lag-Lead compensation



- $\%OS = 10$
- $T_p = 1.61$s.

Ramp response in Matlab: Lag-Lead compensation



- $T_p = 1.61$s.
- $e_{ss} = 0.084$s.

Example 4: PID control from Lag-Lead Compensation

A lag-lead compensator $G_{\text{Lag-Lead}}$ in Example 3 has been design in order to yield a 13.25% overshoot, peak time of 2 seconds and $K_v = 12$ when controlling the following system:

$$G(s) = \frac{K}{s\,(s+1)\,(s+4)}$$

The lag-lead compensator is:

$$G_{\text{Lag-Lead}} = 48\left(\frac{s+0.183}{s+0.0172}\right)\left(\frac{s+0.56}{s+5.96}\right)$$

- Using the designed lag-lead compensator $G_{\text{Lag-Lead}}$ above, design a PID controller that will satisfy the same requirements. Write PID controller in the following form and calculate the values of K_p, T_i, T_d.

$$PID(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

- Compare the lag-lead compensator and the PID controller using a simulation with a step input and noise with frequency 20 rad/s in the measurement.

Example 4: Find PID gains from Lag-Lead Compensation

Use PI part as lag compensator and PD part as lead compensator. Keeping only middle break frequencies $\frac{1}{T_2} = 0.183$ and $\frac{1}{T_1} = 0.56$ with same asymptote between them. The PID is

$$
\begin{aligned}
PID(s) &= \frac{48}{5.96}\frac{(s+0.183)(s+0.56)}{s} = 8.054\frac{0.1025+0.743s+s^2}{s} \\
&= 0.8255\left(\frac{1+7.2488s+9.7561s^2}{s}\right) \\
&= K_p\left(1+\frac{1}{T_i s}+T_d s\right) = \frac{K_p}{T_i}\left(\frac{1+T_i s+T_d T_i s^2}{s}\right) \\
\Rightarrow &\begin{cases} T_i = 7.2488 \\ T_d T_i = 9.7561 \quad \Rightarrow \quad T_d = 1.3463 \\ \frac{K_p}{T_i} = 0.8255 \quad \Rightarrow \quad K_p = 5.9839 \end{cases}
\end{aligned}
$$

Example 4: Alternative method to calculate PID gains from Lag-Lead

Keeping the two middle break frequencies of the lag-lead compensator, the PID controller is:

$$PID(s) = K_1 \frac{(s + 0.183)(s + 0.56)}{s}$$

where K_1 is a constant to be determined.

- The asymptote of the lead-lag compensator between the two middle break frequencies has amplitude:

$$M_a = 48 \frac{0.56}{5.96} = 4.5101$$

- The amplitude of the PID between the two middle break frequencies has amplitude:

$$M_b = 0.56\, K_1$$

- Choosing $M_a = M_b$ the gain K_1 is found:

$$0.56\, K_1 = 4.5101 \quad \Rightarrow \quad K_1 = 8.054$$

Example 4: Alternative method to calculate PID gains from Lag-Lead

Expanding the numerator, the PID is:

$$PID(s) = 8.054\,\frac{0.1025 + 0.743s + s^2}{s} = 0.8255\left(\frac{1 + 7.2488s + 9.7561s^2}{s}\right)$$

Identifying the coefficients with the general form for a PID controller in function of K, T_d and T_i gives:

$$PID(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right) = K_p\left(\frac{T_i s + 1 + T_d T_i s^2}{T_i s}\right)$$

$$\Rightarrow \begin{cases} T_i = 7.2488 \\ T_d T_i = 9.7561 \quad \Rightarrow \quad T_d = 1.3463 \\ \frac{K_p}{T_i} = 0.8255 \quad \Rightarrow \quad K_p = 5.9839 \end{cases}$$

Example 4: Comparison between PID and Lag-Lead

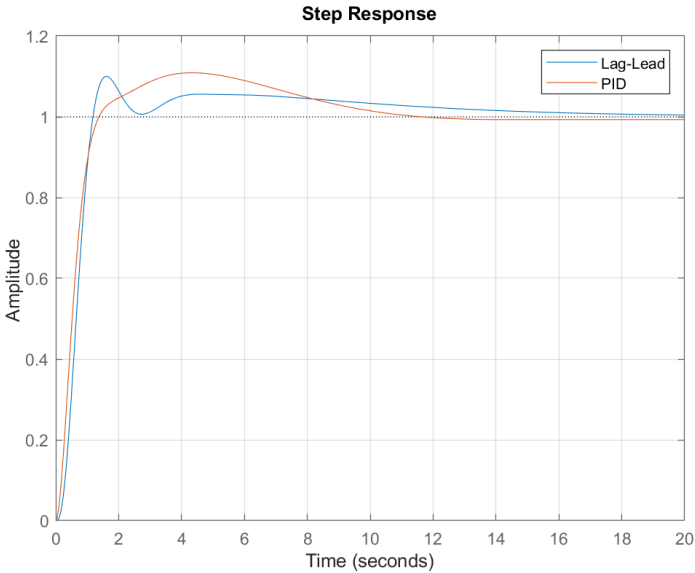


Figure: Comparison between PID and lag-lead compensator